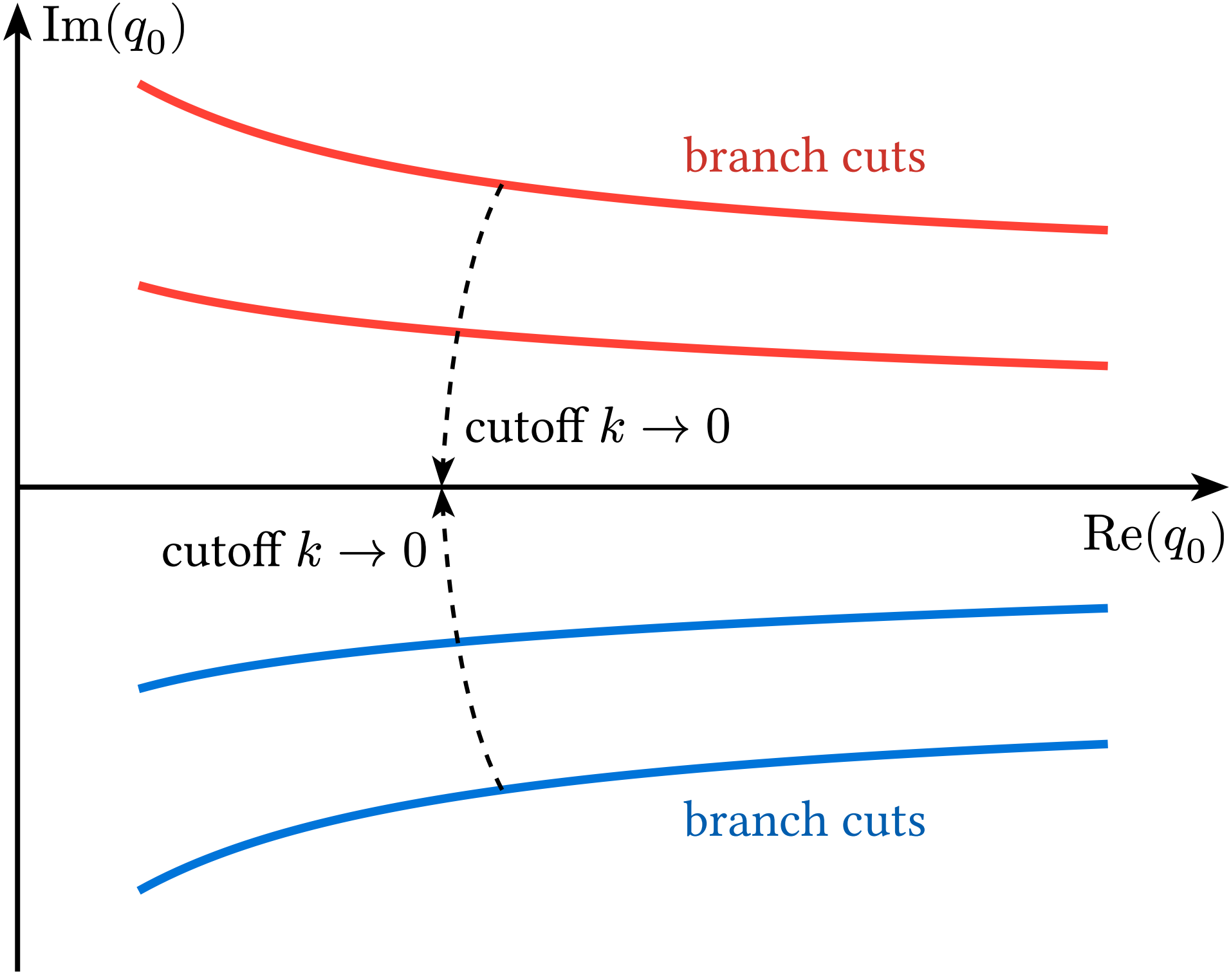
$\mathrm{Im}(q_0)$
branch cuts
cutoff $k \to 0$
cutoff $k \to 0$
$\mathrm{Re}(q_0)$
branch cuts